21 Random Lemmas

Suppose T is tame. Recall that $\Xi(T)$ is (coarsely) defined if $T \searrow \overline{T}$ where $\overline{T} \in FC'$ (the subspace of FC of trees without filling vertex stabilizers).

⟨l:1-iso⟩ **Lemma 21.1.** *Suppose $f : T_1 \to T_2$ is an optimal (correct term?) morphism of tame trees. If T_2 has a point x_2 in the interior of a $\mathbb{1}$-edge such that $f^{-1}(x_2)$ consists of a single point x_1 that is the in the interior of a $\mathbb{1}$-edge, then T_1 and T_2 have a common collapse $\overline{T} \in FS$.*

Proof. Choose a small regular neighborhood I_2 about x_2 so that $f^{-1}(I_2)$ is a small regular neighborhood I_1 about x_1. In particular, f restricts to an isometry $I_1 \to I_2$. Let $\overline{T}$ be the result of collapsing the components of the complement of the orbit of I_i in T_i. □

Lemma 21.2. *Suppose $T_1 \to T_2$ is an optimal morphism of tame trees in FC' that only folds $\mathbb{Z}$-edgelets. Then $\Xi(T_1)$ and $\Xi(T_2)$ have a common collapse.*

Proof. If T_1 (equivalently T_2) has a $\mathbb{1}$-edge then we are done by Lemma 21.1. So suppose T_1 and T_2 have only $\mathbb{Z}$-edges. Let $\widehat{T}_1$ be the result of blowing up a vertex of T_1 to have a $\mathbb{1}$-edge. A folding sequence from T_1 to T_2 lifts to one from $\widehat{T}_1$ to $\widehat{T}_2$. By Lemma 21.1, $\widehat{T}_1$ and $\widehat{T}_2$ have a common collapse. □

The next lemma shows that FS_V is well-defined. Let $S_1, S_2 \in FC$ respectively have the witness V as a the stabilizer of a vertex v_i, let $T_V \in FS_V$, and let $\widehat{S}_i$ be the result of using T_V to blowup v_i.

Lemma 21.3. *$\widehat{S}_1$ and $\widehat{S}_2$ have a common collapse in FS.*

Proof. $X := X(S_1, S_2) \in FC$ and has V as the stabilizer of a vertex x. Also $X \searrow S_1$ and there is a morphism $X \to S_2$. Let $\widehat{X}$ be the result of using T_V to blowup x. There is an induced collapse $\widehat{X} \to \widehat{S}_1$ (and so $\Xi(\widehat{X}) = \Xi(\widehat{S}_1)$. There is also an induced morphism $\widehat{X} \to \widehat{X}_2$. The morphism $\widehat{X} \to \widehat{S}_2$ only folds $\mathbb{Z}$-edgelets. By Lemma 21.1, $\Xi(\widehat{X})$ and $\Xi(\widehat{S}_2)$ have a common collapse in FS. □

Remark 21.4. Similar reasoning shows that the results of collapsing the $\mathbb{Z}$-edges of $\widehat{S}_1$, $\widehat{S}_2$, and $\widehat{X}$ are all equal.

⟨l:quasigeo⟩ **Lemma 21.5.** *if T_t is a folding path in $\overline{\mathcal{O}}(\mathbb{F}, \mathcal{F})$ where each T_t has a $\mathbb{1}$-edge, then $\Psi(T_t)$ is a uniform unparametrized quasigeodesic in FS.* grep from earlier version

⟨l:inj implies inj⟩ **Lemma 21.6.** *Suppose we have a commutative diagram*

$$\begin{tikzcd} T_1 \arrow[r] \arrow[d] & T_2 \arrow[d] \\ \overline{T}_1 \arrow[r] & \overline{T}_2 \end{tikzcd}$$

with horizontal maps morphisms and vertical maps collapses. Choose infinite cyclic $Z < \mathbb{F}$*.*

- *If* $Fix_Z(\overline{T}_i)$ *is a nondegenerate arc then* $Fix_Z(T_i) \to Fix_Z(\overline{T}_i)$ *is onto.*
- *If* $Fix_Z(T_1) \to Fix_Z(T_2)$ *is injective then* $Fix_Z(\overline{T}_1) \to Fix_Z(\overline{T}_2)$ *is injective.*

Proof.

- Suppose $\overline{x} \in Fix_Z(\overline{T}_i)$. The preimage of $\overline{x}$ in T_i is a Z-tree. If there is a Z-fixed point in this preimage then it is in $Fix_Z(T_i)$ and projects to $\overline{x}$. Otherwise, Z acts hyperbolically on T_i with axis in the preimage of $\overline{x}$. It follows that $Fix_Z(\overline{T}_i)$ is the single point $\overline{x}$, contradiciton.
- Suppose not. Let $\overline{x}_1 \neq \overline{x}_1'$ be in $Fix_Z(\overline{T}_1)$ with images equal to $\overline{x}_2 \in \overline{T}_2$. Let $x_1, x_1' \in Fix_Z(T_1)$ respectively map to $x_2, x_2' \in T_2$. (They exist by the first bulleted item.) Since the induced morphism $Fix_Z(T_1) \to Fix_Z(T_2)$ is injective, The image of $[x_1, x_1']$ in T_2 is the arc between the images of x_1 and x_1', which is in the fiber over $\overline{x}$. It follows that the image of the interval $[\overline{x}_1, \overline{x}_1']$ in $\overline{T}_2$ is $\overline{x}$, contradiciton.

□

Lemma 21.7. *Suppose*

- T_t*,* $0 \leq t \leq \omega$*, is the canonical optimal folding path guided by* $T_0 \to T_\omega$*;*
- $T_0, T_\omega \in \overline{\mathcal{O}}$*; and*
- $F_E(T_0) \to Fix_E(T_\omega)$ *is injective for all maximal cyclic* E*.*

Then $T_t \in \overline{\mathcal{O}}$*,* $0 \leq t \leq \omega$*.*

Proof idea. Set $t_0 := \inf\{t | Fix_E(T_t) \to Fix_E(T_\omega) \text{ is not injective}\}$ is zero. Let $\overline{T}_t$ be obtained from of T_t by collapsing the dense Levitt components. There is an induced *generalized folding path* $\overline{T}_t$. Each $\overline{T}_s \to \overline{T}_t$, $s \leq t$

is a collapse followed by a morphism. If s and t are close than the the subsets of $\overline{T}_s$ that are collapses are initial segments of $\mathbb{1}$-edges. Note that $t_0 = \inf\{t | Fix_E(\overline{T}_t) \to Fix_E(\overline{T}_\omega)$ is not injective$\}$. More words.

□

21.1 Substitutes

(Look up Camille's term.)

⟨s:substitutes⟩ *Definition* 21.8. Let T be a tame $(\mathbb{F}, \mathcal{F})$-tree.

- T is *geometric* if it is dual to a lamination on a cocompact two-complex.
- A *substitute for* T is a (simplicial) tree $T' \in FC$ such that if an element of $\mathbb{F}$ is elliptic in T then it is also elliptic in T'.
- A *good resolution for* T is a morphism $f : T' \to T$ where T' is a geometric (not necessarily simplicial) tame $(\mathbb{F}, \mathcal{F})$-tree and T and T' have the same sets of elliptic elements.

Lemma 21.9. *A tame* $(\mathbb{F}, \mathcal{F})$*-tree* T *has a good resolution.*

Proof. Start with any resolution with domain in FS and kill generators of ellipitc subgroups of T by adding two cells. □

⟨l:substitute⟩ **Lemma 21.10.** *Let* T *be a tame* $(\mathbb{F}, \mathcal{F})$*-tree and let* $f : T' \to T$ *be a good resolution.*

- *If* T' *has a* $\mathbb{1}$*-edge or a thin component then* T *has a substitute with a single orbit of edges and all edges are* $\mathbb{1}$*-edges.*
- *If* T' *has neither a* $\mathbb{1}$*-edge nor a thin component then* T' *has surface-type. In fact,* T *has surface-type.*

Proof. Rips theory. □

⟨l:surface-type types⟩ **Lemma 21.11.** *Suppose* T *is a surface-type graph of actions with underlying Bass-Serre presentation* S*.*

(1) *If some vertex tree has filling stabilizer then the vertex tree is single surface with a free (give definition) boundary component.*

(2) *If not (1) and if some* $V \in \mathbf{W}$ *is visible in* S *where* $V|T$ *has dense orbits then the tree* $\overline{T}$ *obtained from* T *by collapsing all simplicial edges of* T *is not a black hole.*

Proof. (1): Outer limits.

(2): Suppose not. Apply Lemma 21.12 where S_1 is the collapse of edges in the subgraph representing V and S_2 is the result of collapses edges not in this subgraph. By assumption, S_2 has a filling vertex stabilizer W. Intersections of conjugates of V and W are vertex groups in the subgraph but these are not filling, contradiction. □

⟨l:two collapses⟩ **Lemma 21.12.** *Suppose $S \in FC$ and $S_1 \swarrow S \searrow S_2$ where each S_i has a vertex v_i with filling stabilizer V_i. Then some conjugates of V_1 and V_2 have intersection that is a witness.*

Proof. It is enough to assume $S = S_1 + S_2$ and show that S has a filling vertex stabilizer. Suppose not. Then distinct vertices of the Bass-Serre diagram for S have $\mathbb{1}$-blowups (needs reference). These vertices must collapse to v_1 and to v_2. This implies that our vertices in S are not distinct, contradiction. (Buy this?) □

⟨p:dense collapse⟩ **Proposition 21.13.** *Suppose T is a tame $(\mathbb{F}, \mathcal{F})$-tree and not a black hole. Either $\Xi(T)$ is defined or the result $\overline{T}$ of collapsing all simplicial edges of T is not a black hole.*

Proof. Suppose $\Xi(T)$ is not defined and that T doesn't have dense orbits. In particular, T has no $\mathbb{1}$-edges and there is a witness $V \in \mathbf{W}$ such that $V|T$ has dense orbits and is a vertex tree in the Levitt graph of actions for T. Replace T by the graph of actions resulting from collapsing the other vertex trees to points (still called T). Since the new graph of actionss still has the original $V|T$ as a vertex tree, the new T is still not a black hole.

If $V|T$ has a replacement T_V with a $\mathbb{1}$-edge, then the conclusion of the proposition holds. Indeed, the result T' of replacing $V|T$ with T_V in T and then collapsing all the original simplicial edges has elliptic set containing that of $\overline{T}$. Since T' also has a $\mathbb{1}$-edge, none of the point stabilizers in question are filling.

Hence, by Lemma 21.10, we may assume that $V|T$ has surface-type. If $V|T$ consists of a single surface, then Lemma 21.11(1) implies that $V|T$ has a free boundary component and a properly embedded arc in the surface with endpoints in this boundary component gives a $\mathbb{1}$-splitting with a single orbit edges and with elliptic set containing all the elliptics of $\overline{T}$ except the subgroup Z represented by the boundary. Z is a vertex stabilizer in $\overline{T}$ and in T. In particular, Z is not filling.

The only remaining case follows from Lemma 21.11(2). □